



Customer Shopping Behavior — Snapshot

Analysis of 3,900 purchases to reveal spending, segments, product prefs, subscriptions.

Dataset Summary



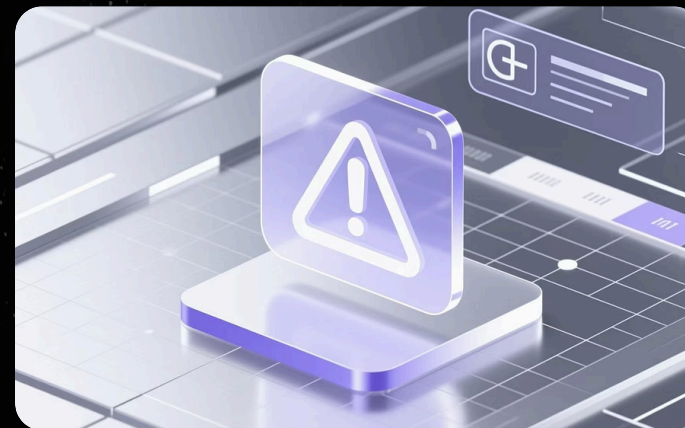
Scale

Rows: 3,900 • Cols: 18



Key Features

Age, Gender, Location, Subscription, Item,
Category, Amount, Season



Missing Data

37 nulls in Review Rating

Data Prep & Engineering

01

Load & Inspect

pandas df.info() and .describe()

02

Clean & Impute

Review Rating imputed with category
median

03

Standardize

Columns → snake_case

04

Feature Engineering

age_group, purchase_frequency_days

05

DB Integration

Loaded cleaned DataFrame into PostgreSQL

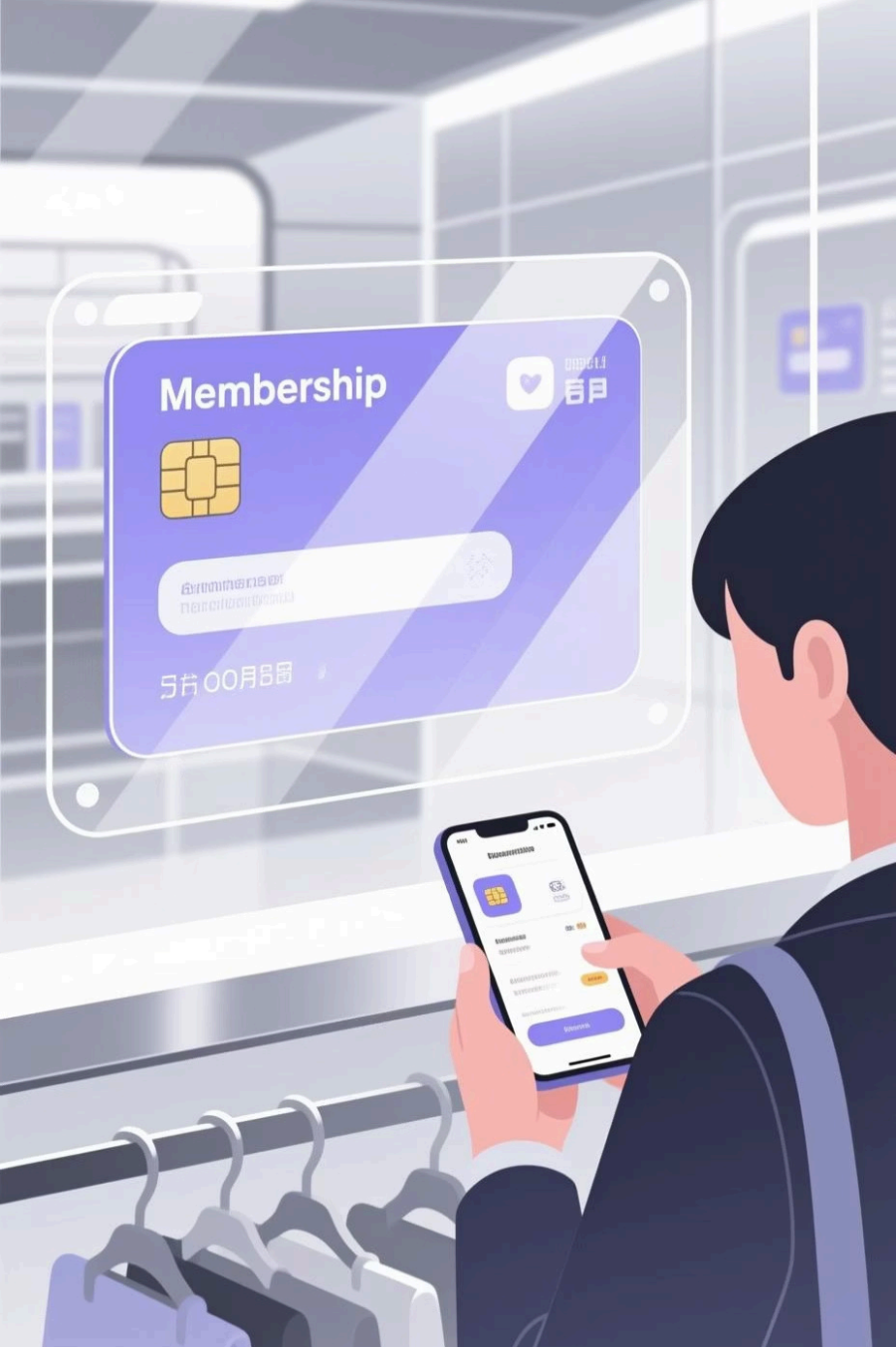
Revenue by Gender

Totals

Male revenue: 157,890

Female revenue: 75,191





Subscriber vs Non-Subscriber Spend

1053

Subscribers

Avg spend: 59.49 • Total revenue:
62,645

2847

Non-Subscribers

Avg spend: 59.87 • Total revenue:
170,436

Top Products & Ratings

Top Rated Items

- Gloves — 3.86
- Sandals — 3.84
- Boots — 3.82
- Hat — 3.80
- Skirt — 3.78





Discounts & High-Spenders

1

High-Spending Discount Users

839 rows identified — users who used discounts yet spent above avg.

2

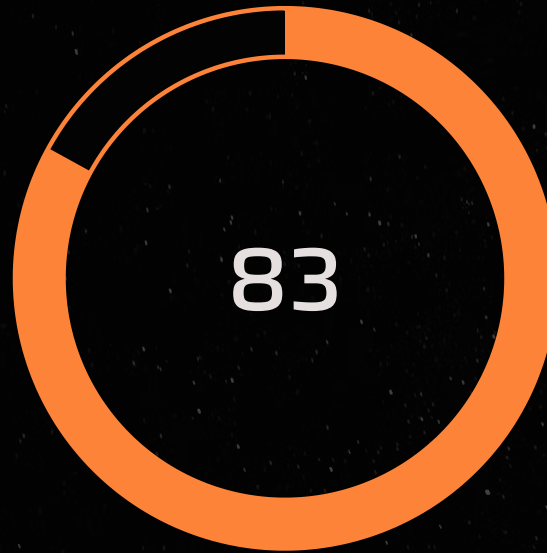
Top Discounted Items

Hat 50% • Sneakers 49.66% • Coat 49.07% • Sweater 48.17% • Pants 47.37%

Customer Segments & Behavior



Loyal



New



Returning

Repeat buyers (>5 purchases) more likely to NOT subscribe: No 2,518 vs Yes 958

Shipping, Payment & Age Insights

Shipping

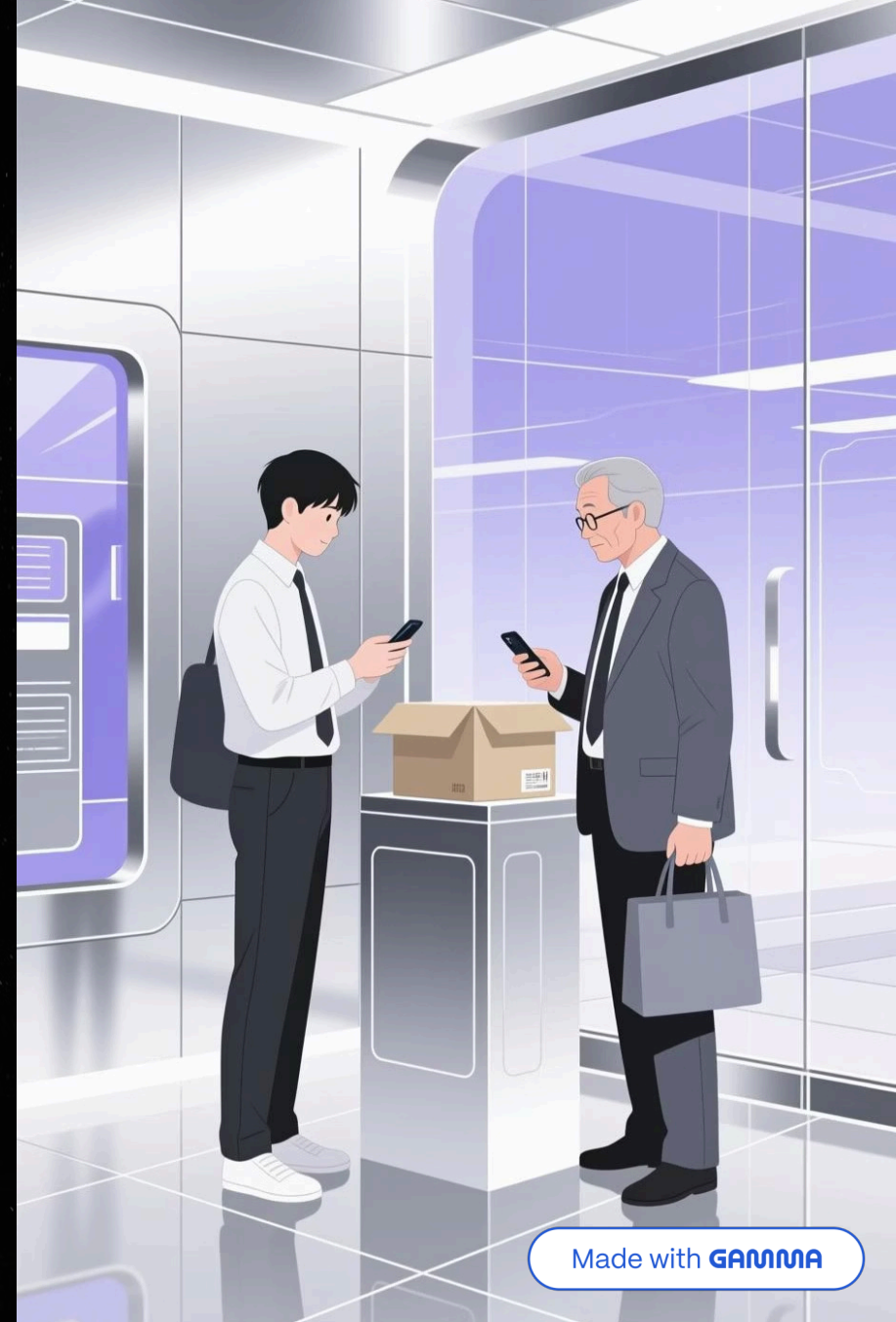
Standard avg: 58.46 • Express avg: 60.48

Payment

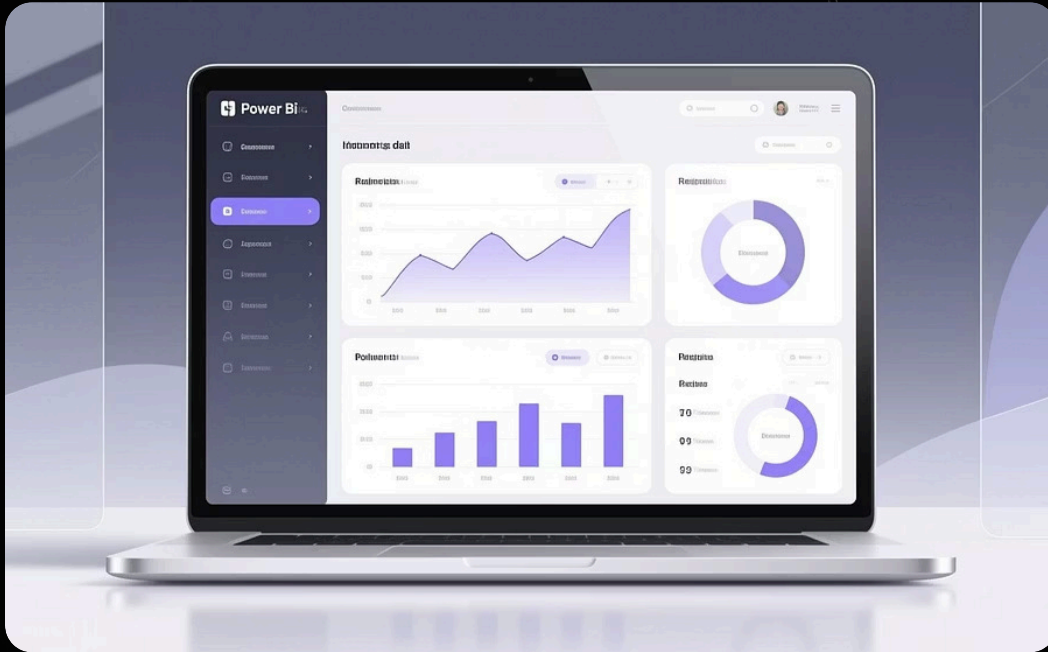
Common method: PayPal

Revenue by Age Group

- Young Adult: 62,143
- Middle-aged: 59,197
- Adult: 55,978
- Senior: 55,763



Recommendations & Dashboard



Dashboard

Interactive Power BI visualizations built for stakeholders.



Recommendations

Boost subscriptions • Reward repeat buyers • Review discount policy • Promote top-rated products • Target high-revenue age groups & express users